pubmed_25429902 ? monitoring

Immune monitoring (IM) has not been shown to be associated with cardiac allograft vasculopathy (CAV). Maximal intimal area, average percent stenosis, plaque volume, and maximal intimal thickness (MIT) were measured for matched baseline and one-yr IVUS segments in a blinded fashion. Patients were divided into quartiles by IM scores and outcomes compared. Optimal IM cutoff was determined. IM assays were measured at 63.7 ± 16.4 d after transplantation in fifty patients. Progression of maximal intimal area ($p = 0.005$), average percent stenosis ($p < 0.001$), plaque volume ($p = 0.005$), and MIT ($p = 0.001$) were increased across the quartiles. An optimal IM assay cutoff of 406.0 ng ATP/mL demonstrated a sensitivity of 66.7% and specificity of 94.3% for predicting rapid progression of MIT ? 0.5 mm. Mean IM scores for Group 1 vs. Group 2 were 176.4 ± 102.2 and 616.3 ± 239.5 ngATP/mL, respectively. Rapid progression of MIT ? 0.5 mm occurred in 5/38 patients (13.2%) in Group 1 vs. 10/12 patients (83.3%) in Group 2 ($p < 0.001$). The risk ratio for rapid progression with elevated IM was 11.7 ($p < 0.001$).

Question: Is elevated immune monitoring early after cardiac transplantation associated with increased plaque progression by intravascular ultrasound?

Answer: Elevated early IM scores are associated with progression of CAV by IVUS. These findings suggest the potential of IM for tailoring of immunosuppressive regimens to minimize the progression of CAV in high-risk patients.